MotoGP PDF Timing Analyzer
Extract lap-by-lap MotoGP timing data from the official MotoGP Chronological Analysis PDF and generate:

- JSON dataset
- Excel workbook
- Individual rider tabs
- Interactive race comparison chart
- Rider visibility checkboxes
- Lap-time analysis

Features
PDF Extraction
Parses official MotoGP Chronological Analysis PDFs.
Extracts:

- Rider name
- Rider number
- Finishing position
- Team
- Manufacturer
- Total laps
- Valid laps
- Lap number
- Lap time
- Sector times (T1-T4)
- Speed
- Cancelled laps
- Pit lane indicators

JSON Output
Creates a structured JSON file containing all riders and lap data.
Example:

```
{
 "name": "Marco BEZZECCHI",
 "number": 72,
 "position": 1,
 "laps": [
  {
   "lap": 1,
   "lap_time": "1:34.554",
   "lap_seconds": 94.554
  }
 ]
}
```

Excel Output
Creates:
Rider Sheets
One worksheet per rider.
Example:

- Marco Bezzecchi
- Raul Fernandez
- Jorge Martin
- Marc Marquez
- Pedro Acosta
- etc.

Each sheet contains:

Lap, Lap Time, Lap Seconds, T1, T2, T3, T4, Speed, Cumulative

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Race Data Sheet
Contains all riders in a normalized format.
Useful for:

- Pivot tables
- Power BI
- Pandas analysis
- Machine Learning

Race Progress Dashboard
Interactive chart with:

- Checkbox per rider
- Individual colors
- Lap vs Lap Time chart
- Hide/show riders dynamically

Checkbox example:
☑ Marco Bezzecchi
☑ Jorge Martin
☐ Marc Marquez
☐ Pedro Acosta

Requirements
Python 3.10+
Install dependencies:
-pip install pdfplumber xlsxwriter

Project Files
Parser
motogp_parser_all_riders.py
Reads:
Analysis.pdf
Creates:
motogp_all_riders.json

Excel Builder
motogp_excel_builder_all_riders.py
Reads:
motogp_all_riders.json
Creates:
motogp_all_riders.xlsx

Usage
Step 1 - Parse the PDF
python motogp_parser_all_riders.py Analysis.pdf
Or:
python motogp_parser_all_riders.py Analysis.pdf -o motogp_all_riders.json
Expected output:
Wrote motogp_all_riders.json: 21 riders, 478 lap rows

Step 2 - Generate Excel Workbook
python motogp_excel_builder_all_riders.py motogp_all_riders.json
Or:
python motogp_excel_builder_all_riders.py motogp_all_riders.json -o race_analysis.xlsx
Expected output:
Wrote race_analysis.xlsx

Excel Workbook Structure

Rider Tabs

Each rider contains:

Columns

Column	Description
A	Lap
B	Lap Time
C	Lap Seconds
D	T1
E	T2
F	T3
G	T4
H	Speed
I	Cumulative Seconds
J	Excel Time

Excel Time Column

Column J contains: lap_seconds / 86400

This allows Excel charts to display:

1:29.960
1:30.215
1:31.005

instead of:

89.960
90.215
91.005

Dashboard Logic

The dashboard dynamically shows/hides riders using formulas.

Example:

=IF(A4,'Marco Bezzecchi'!J6,NA())

When unchecked:

#N/A

Excel automatically hides the line.

Supported Riders

Current version supports:

1. Marco Bezzecchi
2. Raul Fernandez
3. Jorge Martin
4. Ai Ogura
5. Pedro Acosta
6. Marc Marquez
7. Alex Marquez
8. Fabio Di Giannantonio
9. Fabio Quartararo
10. Enea Bastianini
11. Brad Binder
12. Luca Marini
13. Johann Zarco
14. Franco Morbidelli
15. Alex Rins
16. Diogo Moreira
17. Maverick Viñales
18. Toprak Razgatlioglu
19. Francesco Bagnaia
20. Jack Miller
21. Fermin Aldeguer

Current Limitations

This parser is designed specifically for:
MotoGP Chronological Analysis of Performances PDFs
with the same page layout structure as:
Pertamina Grand Prix of Indonesia
Chronological Analysis of Performances
MotoGP may occasionally change PDF layouts.
If a future season changes the format, the SEGMENTS mapping may need adjustments.

Future Enhancements

Potential improvements:

- Gap to leader analysis
- Position-by-lap graph
- Manufacturer comparison
- Team comparison
- Race pace chart
- Sector pace chart
- Fastest-lap ranking
- Automatic PDF layout discovery
- Sprint race support
- Power BI export

Author

Created as part of a MotoGP race analytics project using Python, PDF extraction, JSON processing, and Excel automation.